

AASHISH MUKUND

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EDUCATION

University of Colorado Boulder (Matriculating)

Boulder, CO

Master of Science in Computer Science (GPA- 3.97/4)

Aug. 2023 – May 2025

Coursework: Datacenter Scale Computing, DAA **TA:** CSCI 2400, CSCI 2700 **RA:** Earth Lab

R.V. College of Engineering

Bengaluru, India

Bachelors in Computer Science and Engineering(GPA- 8.91/10)

Aug. 2016 – May 2020

EXPERIENCE (ORDERED BASED ON RELEVANCE)

Walmart Global Tech

Bengaluru, India

Software Engineer II / Software Engineer III

Aug. 2020-Apr. 2023

- Led the efforts in configuring and solutionizing Splunk forwarder (Fluentd) container across a large scale of virtual machines (5000+) in production, automating the process of failure point detection through log monitoring.
- Reduced turn around time for Data Extraction team by developing an API with CQRS using .NET framework.
- By implementing proof-of-concept (PoC) tests, optimized container resource usage and improved governance.
- Led migration efforts from cloud to on-premise virtual machines, achieving \$1.25M in cloud cost savings.
- Worked with Linux, Azure Cloud and IoT Edge, Docker, Kubernetes to streamline deployment and orchestration.

Earth Lab, CIRES

CU Boulder, CO

Graduate Research Assistant

June 2024-Aug. 2024

- Designed and developed fuel status prediction models (MESMA) for SoCal regions, by leveraging Sentinel-2 satellite data. Curated a custom spectral library from Sentinel-2 pure-pixel data, currently being used by the lab.
- Set up a pipeline on AWS for pre-processing the Sentinel-2 data, aiding in building the custom spectral library.

Image and Video Computing Group

CU Boulder, CO

Research Group Member (Advisor - Dr. Danna Gurari)

Jan. 2024-Dec. 2024

- Led novel efforts to identify how vision-language models reason during incorrect visual evidence on tasks like VQA.
- Benchmarked proprietary and non-proprietary models, showing performance gaps in handling question ambiguity.

Indian Institute of Science (IISc RBCCPS)

Bengaluru, India

Research Intern (Advisor - Dr. Raghu Krishnapuram)

Dec. 2019-June 2020

- Generated RGB-D dataset by simulating turtlebot3 using ROS and Gazebo for 3D reconstruction.
- Leveraged the dataset to perform tasks like 3D reconstruction using Microsoft Kinect Fusion, 3D Segmentation of the point cloud using PointNet++ and, scene completion from an incomplete point cloud using ScanComplete.

PUBLICATIONS

Keerthan, **Mukund A.**, Nagaraj, Prakash, "U-shaped Transformers for 3D Lung Cancer Segmentation", International Conference on Knowledge Engineering and Communication Systems 2022, **IEEE** [\[link\]](#)

Keerthan, **Mukund A.**, Nagaraj, Prakash, "LeafViT: Vision Transformers based Leaf Disease Detection", International Conference on Computational Intelligence and Computer Vision 2022, **Springer** [\[link\]](#)

PROJECTS

News Summarisation using ReactJS, GCP and Flask (DCSC Final Project)

Aug. 2024-Dec. 2024

- Designed a news summarization website(frontend-ReactJS) designed to deliver concise news updates efficiently
- Integrates with GCP SQL database, news data retrieved via News API and stored in S3, backend with Flask

Attention Visualization as Evidence For VQA (Independent Study)

Jul. 2024-Dec. 2024

- Implemented a Grad-CAM based approach for visualizing attention in Visual Question Answering.
- Deployed backend on Hugging Face Spaces (Gradio app) and hosted frontend (HTML/CSS/JS) on GitHub Pages.

TECHNICAL SKILLS

Languages: Python, C++, MATLAB, SQL, Java **AI/ML Frameworks:** PyTorch, TensorFlow, Keras, OpenCV
Web & Backend: Flask, Spring Boot, REST API **Cloud & DevOps:** Docker, Kubernetes, AWS, GCP, Ansible
Developer Tools: Git, Visual Studio, Postman, Jupyter Notebook